

SUPPLY CHAIN & COMMERCIAL DIAGNOSTIC

DataCo Global Supply Chain: Where Value Concentrates, Where It Leaks, and What to Do Next

*A data-driven review of 180,519 order transactions (2015–2018)
across 5 markets, 164 countries, and 118 SKUs*

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EXECUTIVE SUMMARY

Revenue Is Healthy but Fragile — One SKU, a Handful of Markets, and a Fifth of All Orders Are Doing the Damage

DataCo Global's order book is large, global, and outwardly stable: \$36.8M in analyzed sales across 180,519 order lines, 65,752 orders, and 20,652 customers spanning five markets and 164 countries. But beneath that scale, four structural issues concentrate risk and quietly erode margin. None of them is visible from top-line sales alone — each only surfaces once the data is decomposed.

- 01 Concentration risk.** One product supplies ~19% of total revenue; just 4 of 50 categories account for 52% of sales. A single-SKU disruption is a company-level revenue event, not a product-line event.
- 02 Margin leakage.** Sales and profit move almost independently ($r=0.13$), while discount depth tracks sales closely ($r=0.62$). Nearly 1 in 5 orders (33,784) lose money outright.
- 03 Delivery reliability.** 55% of orders are logged as "late." Part of this is structural: the company's own scheduled promise undershoots actual delivery time by ~0.6 days on average.
- 04 Risk blind spot.** Suspected-fraud flags track order volume almost exactly by market, shipping mode, and customer segment — evidence the current rule isn't actually risk-differentiating.

BOTTOM LINE: Rebalance the product portfolio, install margin guardrails on discounting, fix the delivery promise, and modernize fraud detection — a four-part, 12-month value-recovery program detailed in this report.

CONTEXT & SCOPE

Diagnostic Scope: A Global Book of Business Spanning 5 Markets and 164 Countries

180,519

Order line items analyzed

65,752

Unique customer orders

20,652

Unique customers

118 SKUs

Across 50 categories, 11 departments

5 markets

23 regions · 164 countries

Jan 2015–Jan 2018

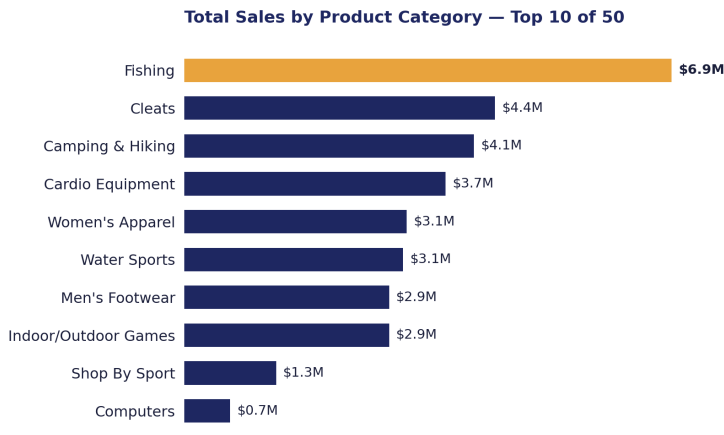
Transaction window (2018 is a partial month)

Methodology: Source data was profiled and cleaned — five unusable fields were removed, including a column that was 100% missing and two constant-value placeholder fields — then enriched with eleven engineered features (shipping duration, profit margin %, delivery performance, and value/sales segmentation) before this analysis was performed. Source: DataCo Smart Supply Chain dataset (Kaggle / DataCo Global).

The remainder of this report walks through five findings, each with the quantified evidence behind it, followed by a synthesis of value at stake, a prioritized set of recommendations, and a proposed implementation roadmap.

FINDING 1 — PORTFOLIO CONCENTRATION

A Single SKU Generates as Much Revenue as the Next Six Product Categories Combined



Total sales by product category, top 10 of 50. The Fishing category is dominated by a single SKU — the Field & Stream Sportsman 16-Gun Fire Safe.

This is a classic Pareto pattern taken to an extreme: revenue depends disproportionately on a handful of products rather than a diversified catalog. A single-SKU stockout, supplier disruption, or demand shift in the Fan Shop department would move company-level revenue, not just a product line's. This concentration is a supply-continuity and category-management issue as much as a sales one — inventory, supplier redundancy, and demand-sensing investment should be weighted toward this narrow set of products accordingly.

18.9% of revenue

comes from a single SKU — more than any full category except its own.

52% of revenue

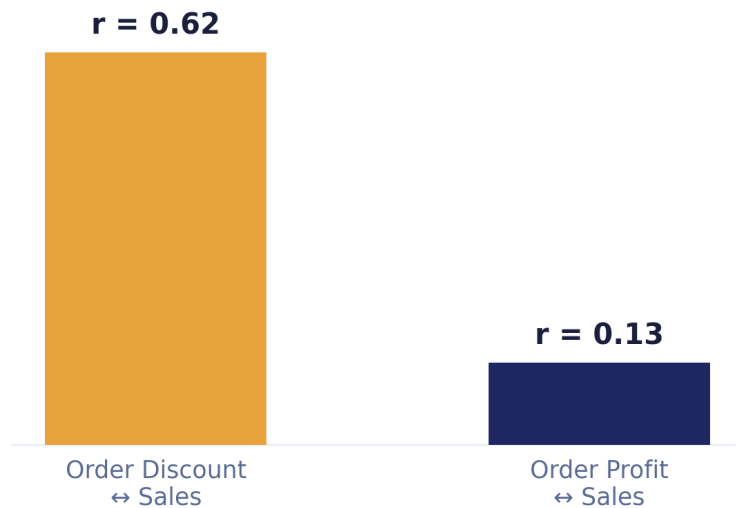
is generated by just 4 of the 50 product categories in the catalog.

46% of revenue

flows through one department (Fan Shop) of 11 — more than double the next-largest, Apparel.

FINDING 2 — MARGIN LEAKAGE

Profitability Is Decoupled from Sales: Discounting Is Buying Revenue, Not Margin



Correlation of order discount and order profit with order sales value, across all 180,519 order lines.

A high correlation between discount depth and order size ($r = 0.62$) shows that discounting is a primary lever behind larger orders — while profit barely follows sales at all ($r = 0.13$). In other words, the business is effectively buying revenue with margin, and the data does not show that trade consistently paying off.

Order-level margin is bimodal: the typical (median) order earns a healthy 27% margin, but a tail of severely negative-margin orders — down to -275% — drags the revenue-weighted blended margin to just ~11%. The problem is not typical performance; it is a subset of value-destroying transactions.

18.7% of orders

(33,784 of 180,519) post a net loss — nearly 1 in 5 transactions.

~\$6.9M in revenue

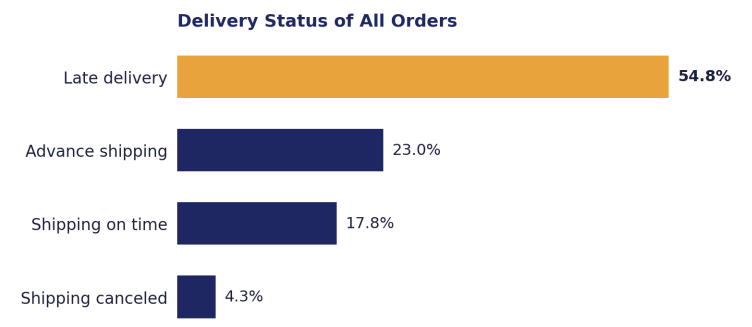
(19% of total sales) is tied up in loss-making orders, estimated from average order value.

-275% margin

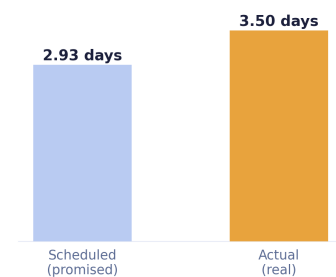
in the worst individual orders — vs. a median order margin of +27%.

FINDING 3 — DELIVERY RELIABILITY

Delivery Reliability Is a Structural Problem, Not Just an Execution One



Delivery status of all 180,519 orders in the dataset.



Scheduled shipping windows average 2.93 days, but actual delivery averages 3.50 days — a 0.57-day (~20%) shortfall against the company's own promise. This gap alone can account for a meaningful share of orders landing in the "late" bucket, independent of any change in warehouse or carrier performance.

55% "late"

of all orders are recorded as late delivery — the single largest delivery-status outcome.

+0.57 days

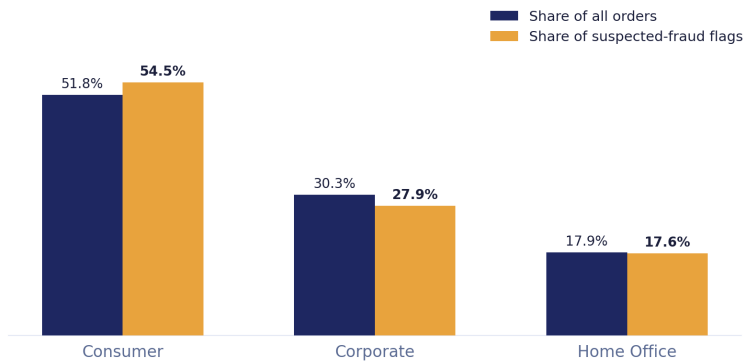
actual shipping exceeds the scheduled promise on average — a ~20% miss against the company's own target.

Only 18%

of orders ship precisely within the promised window — most outcomes fall outside it in either direction.

FINDING 4 — FRAUD RISK BLIND SPOT

Fraud Flags Track Order Volume, Not Risk — the Current Rule May Be a Blunt Instrument



Share of all orders vs. share of suspected-fraud flags, by customer segment.

Roughly 2.25% of orders (~4,000) are flagged as suspected fraud. If the flag captured genuine risk, its distribution should differ from the base rate of ordinary orders — riskier segments, channels, or geographies should be visibly over-represented. Instead, flagged fraud tracks the underlying order mix almost exactly: Consumer is 51.8% of orders and 54.5% of flags; Standard Class shipping is ~60% of orders and ~59% of flags; LATAM is ~28.6% of orders and ~29.1% of flags.

That near-1:1 proportionality across three independent dimensions is the signal: the current flagging approach is most likely a volume-driven heuristic rather than a targeted risk model, and it is probably both under-catching genuine fraud in low-volume corners and over-flagging in high-volume ones.

~4,000 orders

(2.25% of all orders) are currently flagged as suspected fraud.

54.5% vs 51.8%

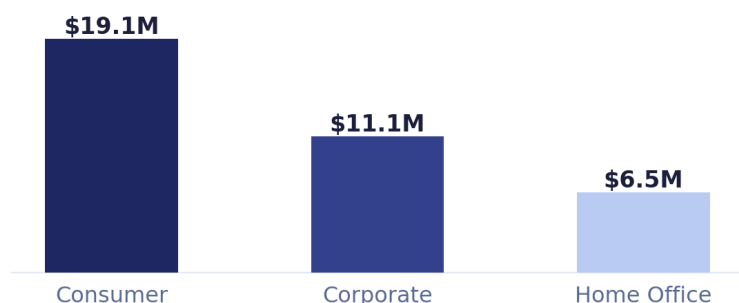
Consumer's share of fraud flags vs. its share of orders — almost no lift.

59% vs 60%

Standard Class's share of fraud flags vs. its share of orders — effectively proportional.

FINDING 5 — CUSTOMER BASE & GEOGRAPHY

The Customer Base Is Broad and Transactional — Not a Small Set of Whales



Total sales by customer segment.

It is tempting to assume a "small number of high-value customers drive revenue" — the underlying notebook's own analysis made that assumption. The data does not support it: across 20,652 customers, the single largest customer accounts for roughly 0.03% of total revenue, and the top 10 combined account for only ~0.25%. With customers averaging 3.2 orders each, this is a broad, transactional base, not a whale-driven one.

The practical implication: a "VIP loyalty program for top customers" will have limited enterprise impact given how thin that concentration is. The more consequential levers are segment-level economics (Consumer 52% of revenue, Corporate 30%, Home Office 18%) and basket-building / cross-sell across the broad base — alongside the sharp geographic concentration below, where the top 10 of 164 countries generate roughly 56% of total revenue.

0.03% of revenue

is the single largest customer's share of \$36.8M total sales.

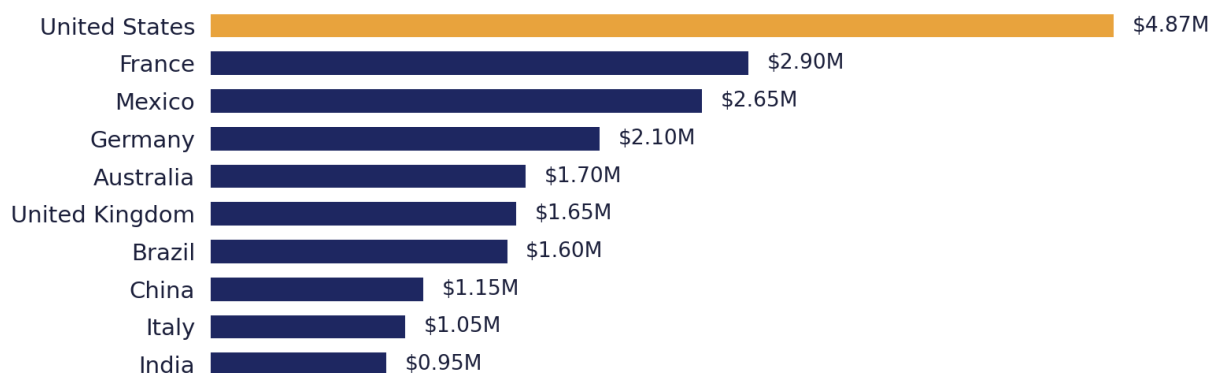
~0.25% of revenue

is the combined share of the top 10 customers out of 20,652.

3.2 orders

is the average number of orders per customer — a transactional, not relationship-driven, base.

Total Sales by Country — Top 10 of 164



VALUE AT STAKE

Four Value Pools, One Integrated Program

The findings above are not independent problems — they compound. A concentrated portfolio makes margin leakage on the top SKUs more consequential; a broken delivery promise erodes the loyalty of exactly the transactional customer base the business depends on; and a fraud rule with no discriminating power leaves both false positives (frustrating good customers) and false negatives (real fraud) on the table simultaneously.

Value Pool	Headline Evidence	Directional Impact	Time to Value
Product & Portfolio	1 SKU = 19% of revenue; 4 categories = 52%	High — continuity risk	3–6 months
Margin & Discounting	18.7% of orders (~\$6.9M) are loss-making	High — direct P&L	0–3 months
Delivery Reliability	55% "late"; 0.57-day promise gap	Medium–High — retention	0–3 months
Fraud & Risk Intelligence	Fraud flag $\approx$ proportional to volume	Medium — loss avoidance	3–6 months

The margin and delivery levers are prioritized first because they are largely within existing operational control and do not require new data infrastructure. Portfolio and fraud-model work take longer because they require supplier/category strategy input and a proper labeled-data build, respectively — both are addressed in the roadmap on the following pages.

RECOMMENDATIONS

A Four-Pillar Action Plan Matched to Each Diagnosed Value Pool

1. Product & Portfolio

- Build supplier and inventory redundancy specifically for the top SKU and top 4 categories; treat single-SKU risk as a continuity issue, not a merchandising one.
- Run an assortment review of the bottom-performing categories (e.g., Computers, Book Shop) for pricing, promotion, or rationalization.

2. Margin & Discounting

- Investigate the 33,784 loss-making orders directly — segment by product, discount band, and shipping cost to isolate the specific driver(s) of negative margin.
- Introduce a margin floor / discount-approval guardrail so high discounts require a minimum-margin check before they're applied.

3. Delivery Reliability

- Re-baseline scheduled shipping windows to reflect actual carrier performance, closing the 0.57-day promise gap at the source.
- Stand up proactive tracking for high-late-delivery-risk shipments (Late_delivery_risk flag) with intervention before the promised date lapses.

4. Fraud & Risk Intelligence

- Replace or supplement the current volume-correlated flag with a proper anomaly-detection model trained on order, customer, and payment-type features.
- Validate the new model by checking that its flag rate is no longer proportional to raw order volume — a direct test of true risk-differentiation.

IMPLEMENTATION ROADMAP

A Phased, 12-Month Path from Quick Wins to Structural Fixes

0–3 months — Quick Wins	3–6 months — Build	6–12 months — Scale
• Audit and act on the loss-making order tail • Introduce discount/margin guardrails • Re-baseline delivery-promise windows • Publish a weekly late-delivery-risk watchlist	• Supplier redundancy plan for top SKU / top 4 categories • Assortment review of underperforming categories • Design and train an anomaly-detection fraud model • Pilot proactive shipment-intervention workflow	• Roll out the fraud model in production, retire the legacy flag • Extend margin guardrails across full catalog / discount workflow • Embed concentration & delivery KPIs into standing executive reporting • Reassess portfolio mix following supplier-redundancy build-out

What's Needed to Move from Diagnosis to Delivery

This diagnostic was built entirely from order-transaction data. Several of the highest-value next steps require data and sponsorship that sit outside this dataset:

- **True cost data (COGS, logistics cost per shipment):** the current "profit" field is a per-order margin proxy; a landed-cost view would sharpen the loss-making-order investigation and make the \$6.9M estimate precise rather than directional.
- **CRM / customer relationship data:** the raw file has no repeat-purchase, lifetime-value, or contact-channel data beyond masked placeholders; integrating actual CRM data would let the customer-strategy reframing in Finding 5 go from directional to actionable.
- **An executive sponsor for the fraud-model workstream:** building and validating a real anomaly-detection model is a 3–6 month, cross-functional (Risk, Data Science, Ops) effort that needs a named owner and access to labeled historical fraud outcomes, not just the "suspected" flag.
- **Category and supplier stakeholder input:** the portfolio-concentration finding is a starting diagnosis, not a merchandising decision — Category Management should weigh in before any supplier-redundancy or assortment-rationalization plan is finalized.

APPENDIX

Data Quality & Methodology Notes

Data cleaning. Five fields were removed prior to analysis: Product Description (100% missing), Order Zipcode (86% missing), and Customer Email / Customer Password / Product Status (constant placeholder values across all 180,519 rows, and in the case of email/password, masked for privacy in the source dataset). No fully duplicate rows were found; 8 records had a missing Customer Last Name and 3 had a missing Customer Zipcode, both immaterial.

Feature engineering. Eleven features were derived to support this analysis: order year/month/quarter/day, weekend-order flag, shipping duration (shipping date minus order date), a high-value order flag (median split), a sales-value tercile category, a profit/loss category, profit margin %, and an engineered "on time vs. delayed" shipping-performance label (validated by cross-tabulation against the source Delivery Status field).

Limitations.

- The dataset spans January 2015 through January 2018; 2018 is a single partial month and is excluded from year-over-year comparisons in this report.
- "Sales" and "Profit" figures are analyzed at the order-line level, consistent with the source data; some totals in this report (e.g., the \$6.9M loss-tied-revenue estimate) are derived using average order value as a proxy rather than a direct per-order sum, and are labeled as estimates accordingly.
- The dataset contains no landed-cost (COGS, freight cost) fields; "profit" reflects the source data's own order-profit calculation, not a fully loaded margin.
- Product catalog is small (118 SKUs), which mechanically produces high category- and SKU-level concentration ratios; this does not diminish the continuity-risk implication but should be read with that scale in mind.
- Suspected-fraud labels reflect the source system's existing flag, not a verified fraud outcome; Finding 4's conclusion is about the flag's statistical behavior, not a claim about true fraud incidence.

Source: DataCo Smart Supply Chain dataset (DataCo Global / Kaggle), 180,519 rows, 53 original columns (61 after feature engineering).